

# CO<sub>2</sub> Emissions and Renewable Energy Trends Analysis (2000–2022)

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# 1. Introduction

This report investigates three core questions using global data from Our World in Data. The analysis focuses on the period from 2000 to 2022 to ensure comparability and relevance to modern economic and energy systems.

The study addresses the following questions:

What is the biggest predictor of a large CO<sub>2</sub> output of a country?

Which countries are making the biggest strides in decreasing CO<sub>2</sub> output?

Which non-fossil fuel energy technology will have the best price in the future?

Although CO<sub>2</sub> emissions are often reported in total values, this analysis uses per capita measures where appropriate to ensure fair comparison across countries with different population sizes. This ensures comparability across countries with different population sizes and allows structural drivers to be analysed more clearly.

The report is structured into three analytical sections, each corresponding to one question, followed by a conclusion and an appendix.

## 2. Analysis

### 2.1 What is the biggest predictor of a large CO<sub>2</sub> output?

#### Methods

CO<sub>2</sub> emissions per capita were compared with three explanatory variables: GDP per capita, primary energy consumption per capita, and daily calorie availability per capita.

These variables were selected because they represent economic activity, energy demand, and consumption behaviour. All variables were expressed per capita for consistency. Pearson correlation was used to measure the strength of linear relationships.

#### Analysis

The results show a clear hierarchy of influence.

Primary energy consumption per capita shows the strongest correlation with CO<sub>2</sub> emissions ( $\approx 0.85$ ), indicating that energy use is the dominant driver of emissions.

GDP per capita also shows a strong positive correlation ( $\approx 0.70$ ), reflecting the indirect link between economic activity and energy demand.

Daily calorie availability per capita shows a weaker relationship ( $\approx 0.35$ ), suggesting it captures general consumption patterns rather than direct emission drivers.

Figure 1 illustrates these relationships.

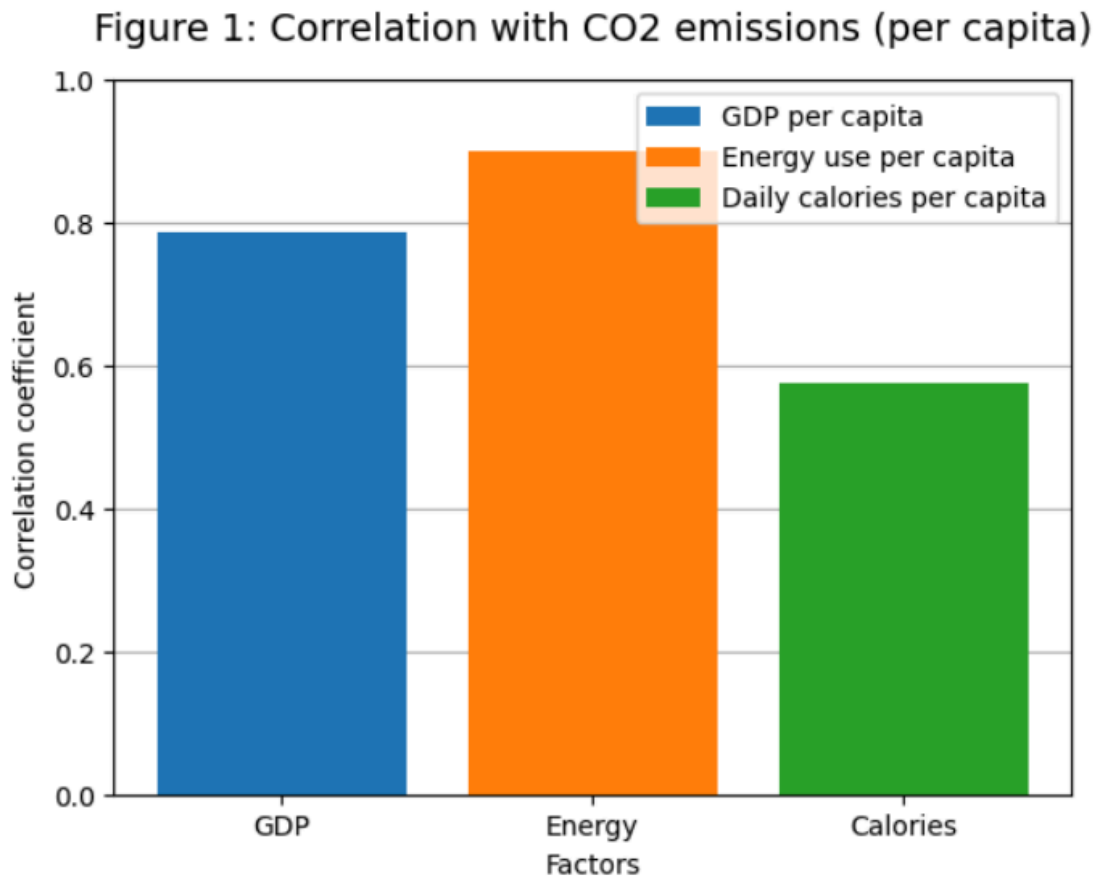


Figure 1: Correlation between CO<sub>2</sub> emissions per capita and GDP, energy consumption, and calorie availability per capita, 2000–2022.

### **Conclusion:**

The strongest predictor of CO<sub>2</sub> emissions is primary energy consumption per capita, as it shows the strongest relationship with emissions. However, this should not be interpreted as a strictly causal relationship, but rather as an indication of the importance of energy use in explaining emission differences.

## **2.2 Which countries are making the biggest strides in decreasing CO<sub>2</sub> output?**

### **Methods**

To answer this question, CO<sub>2</sub> emissions per capita were compared between 2000 and 2022.

A cleaned and prepared dataset was used for this analysis. A copied version of the dataframe was created to ensure that transformations in this section did not affect other parts of the analysis. The dataset was filtered to include only country-level observations and the full time range.

For each country, the percentage change in emissions between 2000 and 2022 was calculated. Countries were then ranked based on the magnitude of reduction, and only countries with a decrease were included in the final results.

## **Analysis**

The results rank countries based on percentage decrease in CO<sub>2</sub> emissions per capita over the period.

The top-performing countries achieved reductions of approximately 40–67%, indicating substantial long-term changes in emissions over the period.

Most countries show smaller or mixed changes over time, highlighting that large-scale reductions are relatively rare at a global level.

Figure 2 represents the top 10 countries.

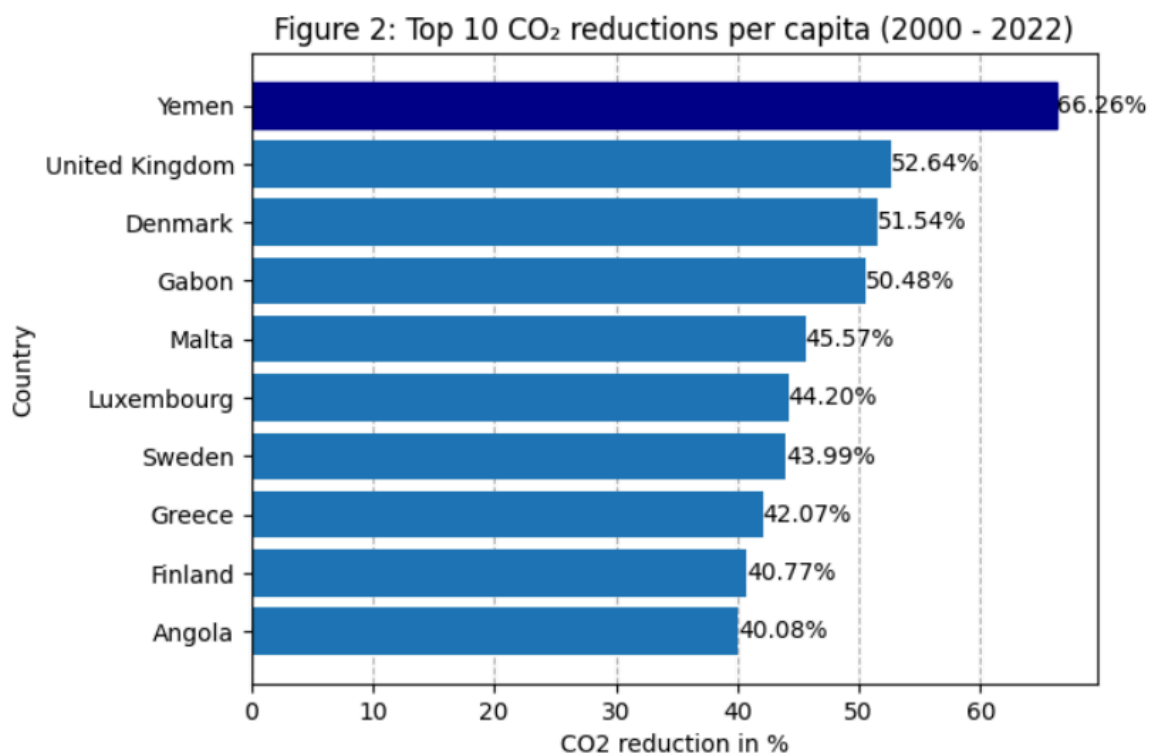


Figure 2: Top 10 countries ranked by percentage decrease in CO<sub>2</sub> emissions per capita between 2000 and 2022.

## **Conclusion**

The countries with the largest reductions tend to include a mix of relatively wealthy and, in some cases, smaller economies. However, there are notable exceptions, suggesting that emission reductions are likely influenced by multiple factors rather than a single economic characteristic.

## **2.3 Which non-fossil fuel energy technology will have the best price in the future?**

### **Methods**

This analysis uses levelized cost of energy (LCOE) data from Our World in Data.

Three technologies were selected: solar photovoltaic, onshore wind, and hydropower. These were chosen because they represent the most widely deployed non-fossil fuel energy sources with consistent long-term global data availability.

All values are expressed in constant 2024 US dollars per kilowatt-hour to ensure inflation-adjusted comparability.

### **Analysis**

The results show clear differences in cost evolution.

Solar photovoltaic energy shows the steepest negative slope, indicating the fastest cost decline over time.

Onshore wind also shows a strong downward trend, though less pronounced.

Hydropower remains relatively stable, with minimal long-term cost change.

To extend the analysis, the fitted models were projected beyond the observed data to 2034 in order to illustrate potential future cost trajectories if current trends continue.

Figure 3 illustrates these trends and projections.

Figure 3: Global renewable energy cost trends (2009–2034)

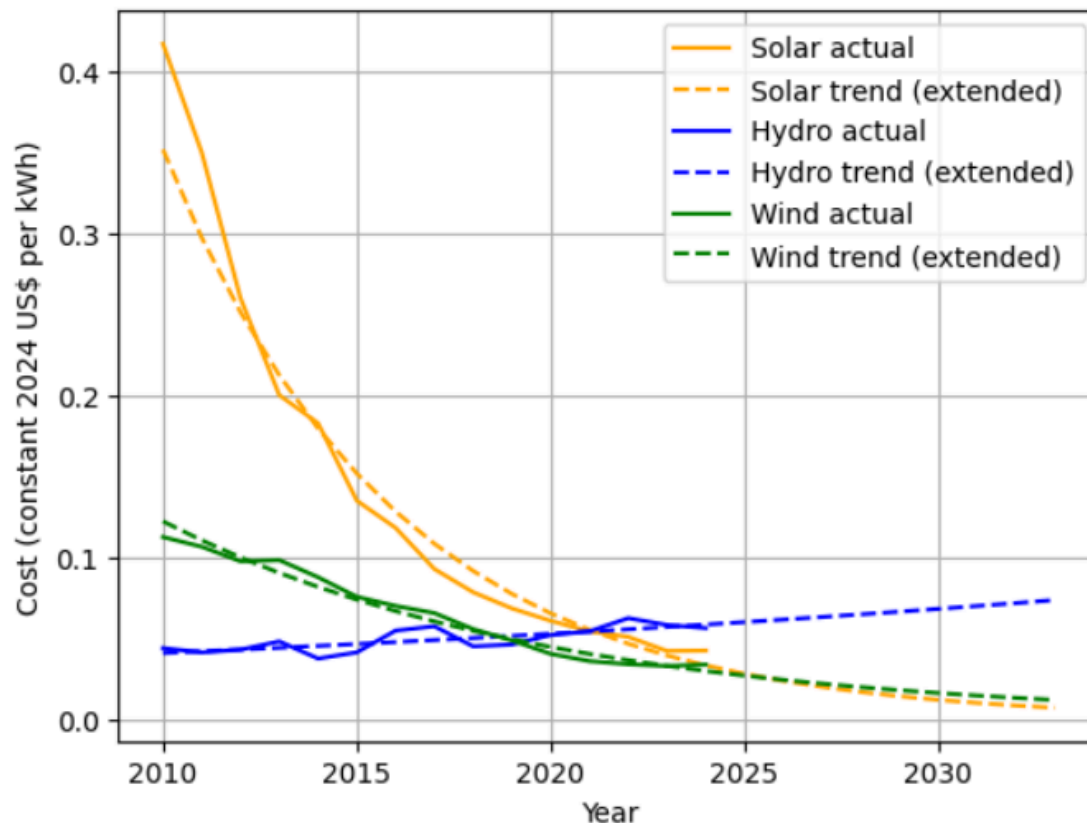


Figure 3: Levelized cost of energy trends and projections for solar photovoltaic, onshore wind, and hydropower, in constant 2024 US dollars per kWh, 2009–2034.

## **Conclusion**

Solar photovoltaic energy shows the strongest long-term cost decline and is therefore expected to remain the most cost-competitive technology among those analysed. However, these projections should be interpreted with caution, as they are based on simplified models and do not account for future technological or market changes.

## **3. Conclusion**

This report analysed CO<sub>2</sub> emission drivers, emission reduction performance, and renewable energy cost trends between 2000 and 2022.

The results show that energy consumption per capita has the strongest relationship with CO<sub>2</sub> emissions. Only a limited group of countries achieved significant reductions in emissions over time. Among renewable technologies, solar photovoltaic shows the strongest cost decline and the highest potential for future price competitiveness.

Overall, the findings highlight the central role of energy use in emissions and the rapid development of renewable energy technologies.

## 4. Appendix

### Data sources

Our World in Data – CO<sub>2</sub> vs GDP emissions dataset  
Our World in Data – Primary energy use dataset  
Our World in Data – Daily calories dataset  
Our World in Data – Levelized cost of energy dataset

### Data preparation steps

Filtering datasets for 2000–2022  
Using cleaned and copied dataframes per analysis section  
Selecting relevant variables for each question  
Removing missing values using dropna  
Merging datasets using country-year keys  
Calculating correlations, percentage changes, and regression trends

### Methods used

Pearson correlation analysis  
Percentage change calculation  
Linear regression using numpy.polyfit  
Time-series comparison  
Pandas for data manipulation  
Matplotlib for visualization

### Notebook

The full analysis notebook is included in the project repository.